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# A causal-guided graph transformer with interpretability analysis for process fault diagnosis

Shuaiyu Zhao, Haoyu Yang, Zihao Wang, Faisal Khan, Qingsheng Wang *

*Mary Kay O'Connor Process Safety Center, Artie McFerrin Department of Chemical Engineering, Texas A&M University, College Station, TX 77843, USA*

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A B S T R A C T

Process fault detection and diagnosis (FDD) is essential for keeping chemical processes safe, but many data-driven models are hard to trust because they often rely on patterns that are not truly related to the real causes of faults. To address this, we introduce a causal-guided graph transformer with dual-layer interpretability analysis (CGTIA). It first builds a directed fault-propagation graph using a lag-aware Granger screening method and then reduces noisy or unnecessary connections with a sparsification strategy. Building on this graph, the model separates learning into two parts: a local branch that follows causal directions and a global branch that captures broader interactions and shared operating conditions. In addition, the method also includes a dual-layer interpretability system that highlights which connections matter most and provides stable, sensor-level explanations. Lastly, we validate the approach on a real multiphase flow facility dataset, which is naturally imbalanced and therefore more challenging than typical simulation datasets. Across various faults, the proposed CGTIA framework achieved a macro averaged recall of 99.20% while maintaining robust variable propagation interpretability. Overall, this combination effectively enhances confidence in the diagnostic results and gives clearer insight into how faults behave in the process.

## 1. Introduction

Process fault detection and diagnosis (FDD) is treated as a first-line safeguard for modern chemical and energy systems, where tight operating margins and complex recycles can convert small deviations into rapid escalation (Kumari et al., 2022a). The latest incident report (Owens et al., 2025a) from the U.S. Chemical Safety Board (CSB) continues to reveal a recurring pattern of potential process risks. Abnormal conditions can often be detected through sensor data long before a final accident occurs (Jiao et al., 2020; Zhang et al., 2026), while alarm systems, human judgment and existing monitoring logic may fail to recognize developing hazards in time. For instance, the steam release incident at Kingsport resulted in $25 million in damages (Owens et al., 2025b), with the root cause identified as the absence of a pipeline condition monitoring program. Also, the Valero port accident in 2023 occurred due to failure to properly identify the risk of hose overpressure, leading to the escape of 6400 pounds of flammable mixture and triggering a major fire. These lessons motivate fault diagnosis models that are not only accurate under realistic disturbances (Bi et al., 2023; Yang et al., 2026), but also transparent enough to support timely intervention and reliable post hoc reasoning.

Following recent advances (Khurshid et al., 2026; C. Wu et al., 2026; Zhao et al., 2026), data-driven fault diagnosis has evolved from feature-engineered classifiers toward deep architectures that learn discriminative representations directly from multivariate time series. Early deep FDD studies typically focused on unsupervised or layer-wise feature learning, for example, Zhang and Zhao (2017) developed a deep belief network-based diagnosis framework to extract compact fault features from nonlinear process data, while later variants (Wang et al., 2020) enhanced this line of work by fusing raw measurements with hidden representations during pretraining to alleviate information attenuation and to better preserve diagnostically useful cues. In parallel, sequence modeling approaches were introduced to explicitly account for process dynamics and fault evolution, where convolutional encoders (Song and Jiang, 2022) were often combined with recurrent units (Arunthavanathan et al., 2021) to capture multi-scale temporal dependencies and to enable early symptom detection or short horizon diagnosis. More recently, attention mechanisms (Jing et al., 2024; Zhao et al., 2024; Chen et al., 2026) and autoencoder structures (Han et al., 2024; Lu and Wang, 2025) have been incorporated to improve feature

* Corresponding author.
*E-mail address:* qwang@tamu.edu (Q. Wang).

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selectivity and provide variable interpretability, typically by emphasizing channels that contribute most to the predicted fault state or by linking abnormal patterns to model outputs. However, although the widespread adoption of pipelines emphasizing temporal modeling and achieving strong performance on curated benchmarks (Kumari et al., 2020; X. Liu et al., 2025), they often treat sensors as an unordered vector and therefore underutilize the process topology that governs how disturbances spread.

Another research thread argues that causality can provide a more appropriate language for fault propagation (Zhu et al., 2026), because industrial disturbances typically manifest through direction dependent mechanisms and time-lagged influence. Many studies (Liu et al., 2023, 2026) have attempted to move beyond the vector formulations by introducing relevance graphs that encode variable structure and enable system reasoning in complex industrial processes. In this line of research, graph neural networks (Ahmad et al., 2024) and attention-based variants are commonly employed, where adjacency is obtained from correlation measures (Luo et al., 2024; Gbashi et al., 2025), mutual information criteria (Ji et al., 2021; Liu et al., 2024) or engineering heuristics (Heimar Andersen et al., 2024; Li et al., 2026), and diagnosis is performed by aggregating neighborhood information to improve discrimination under multivariate coupling. Such relevance-driven methods provide a practical way to incorporate topology, but they often remain undirected and densely connected, making the learned dependencies sensitive to operating mode changes and allowing spurious co-variations to dominate message passing. Motivated by these limitations, an emerging direction is to introduce causal graphs that explicitly model delay, aiming to align diagnostic inference (Zhang and Liu, 2026) with fault propagation mechanisms rather than correlation patterns alone. Here, (Wu et al., 2026) proposed a causal inference framework based on Kolmogorov complexity that can mine historical fault databases and detect fault propagation paths. Importantly, a common experimental practice is to validate these models predominantly on simulation datasets, most notably the Tennessee Eastman Process (TEP) (Qin and Zhao, 2022; Guo and Zhang, 2024), leaving the exploitation of explicit process topology and directional propagation structure comparatively underexplored.

In general, many existing approaches remain difficult to deploy in safety critical settings because their decision logic is opaque (Ewuzie et al., 2025), their learned dependencies are often dominated by non-causal correlations (Dalian et al., 2026), and their explanations are not aligned with physically meaningful fault propagation. Moreover, industrial faults frequently exhibit directional and delayed influence across variables (Kumari et al., 2022b), while widely used graph-based formulations either rely on undirected similarity structures or adopt conventional simulation benchmarks that obscure causal pathways under heterogeneous operating regimes. To bridge these gaps, this paper proposes a cutting-edge **C**ausal-guided **G**raph **T**ransformer with dual-layer **I**nterpretability **A**nalysis (CGTIA) that explicitly integrates the lag-aware causal prior with decoupled representation learning, enabling both reliable diagnosis and mechanism consistent explanations for real industrial processes.

Unlike existing graph transformer-based FDD methods, CGTIA integrates a lag-aware directed causal prior, decoupled local-global reasoning, and dual-layer interpretability for mechanism-consistent diagnosis. Specifically, the principal contributions of this work are summarized as follows.

(i) A Granger lag-aware graph construction strategy is developed to encode directional propagation evidence, and a node-wise sparsification scheme is introduced to suppress noisy dense connections while preserving stable and controllable graph complexity.
(ii) A decoupled causal-context CGTIA architecture is proposed, in which the local branch performs direction-sensitive message passing constrained by the directed topology, while the global branch captures broader system couplings and shared operating conditions that may not be observable through short causal chains.
(iii) A real multiphase flow process facility exhibiting substantial class imbalance is employed, providing a more rigorous evaluation setting than conventional simulation benchmarks and demonstrating the robustness of CGTIA framework.
(iv) A dual-layer interpretability module is established from both decision perspectives: revealing which directed links most influence the prediction, and identifying which sensors consistently contribute to diagnostics, thereby enabling actionable fault propagation insight.

## 2. Methodology

Building on the above motivation, this section details the proposed end-to-end pipeline that transforms raw multivariate sensor streams into an accurate and mechanism consistent fault diagnosis model with transparent explanations, as shown in Fig. 1. We first construct a window level directed sensor graph using lag-aware Granger screening, so the topology can reflect directional and delayed influence rather than undirected similarity. The resulting causal evidence is converted into continuous edge scores and then stabilized via node-wise sparsification, which mitigates dense noise connections while preserving the most credible propagation routes. On this directed prior, the CGTIA encoder adopts a decoupled design: the local causal propagation branch performs direction-sensitive message passing constrained to causal parents, while the global context branch applies graph-wise attention to capture distributed couplings and shared operating conditions beyond short-hop chains. The two branches are fused only after independent normalization and residual refinement, enhancing robustness under heterogeneous regimes and evolving fault stages. Finally, the proposed methodology integrates dual-layer interpretability enabling reliable diagnostics of complex systems and actionable explanatory guidance.

### 2.1. Granger causal-guided graph construction

Given the multivariate nature of complex chemical processes, dynamic coupling relationships are widely observed, and fault propagation inherently exhibits directionality. For this reason, undirected graphs based solely on correlation cannot adequately encode propagation directions. To address this, we construct a directed causal prior model and employ it as a structural inductive bias in the message passing process.

The sensor graph is first defined as:

$$G = (V, E, A), V = s_1, \ldots, s_{24}, A \in \mathbb{R}^{24\times 24} \tag{1}$$

Here, the object manipulated throughout the learning pipeline is specified, with $V$ denoting fixed physical sensing entities, $E$ capturing directional candidate influence paths, and $A$ storing causally filtered edge strengths. In this study, $A$ is not a generic similarity matrix but is explicitly associated with lag-aware Granger evidence, which gives the graph with a process propagation interpretation rather than a purely geometric one.

The mechanism for incorporating temporal signals into the graph model is illustrated in Eq. (2), where each node receives a trajectory of length $T$ instead of a scalar snapshot. For each window sample $w$, the node-feature matrix is:

$$X^{(w)} = \left[x_1^{(w)}, \ldots, x_{24}^{(w)}\right]^\top \in \mathbb{R}^{24\times T}, T = 50 \tag{2}$$

This formulation can address both initial and evolving faults, as the short-term dynamics inside a window contain diagnostic clues that would be lost under static feature summaries. Then, for each ordered pair $(s_i, s_j)$, we estimate whether the lagged $s_i$ improves prediction of $s_j$. At the lag order $L$ (default equal to 5), the constrained and unconstrained models are respectively:

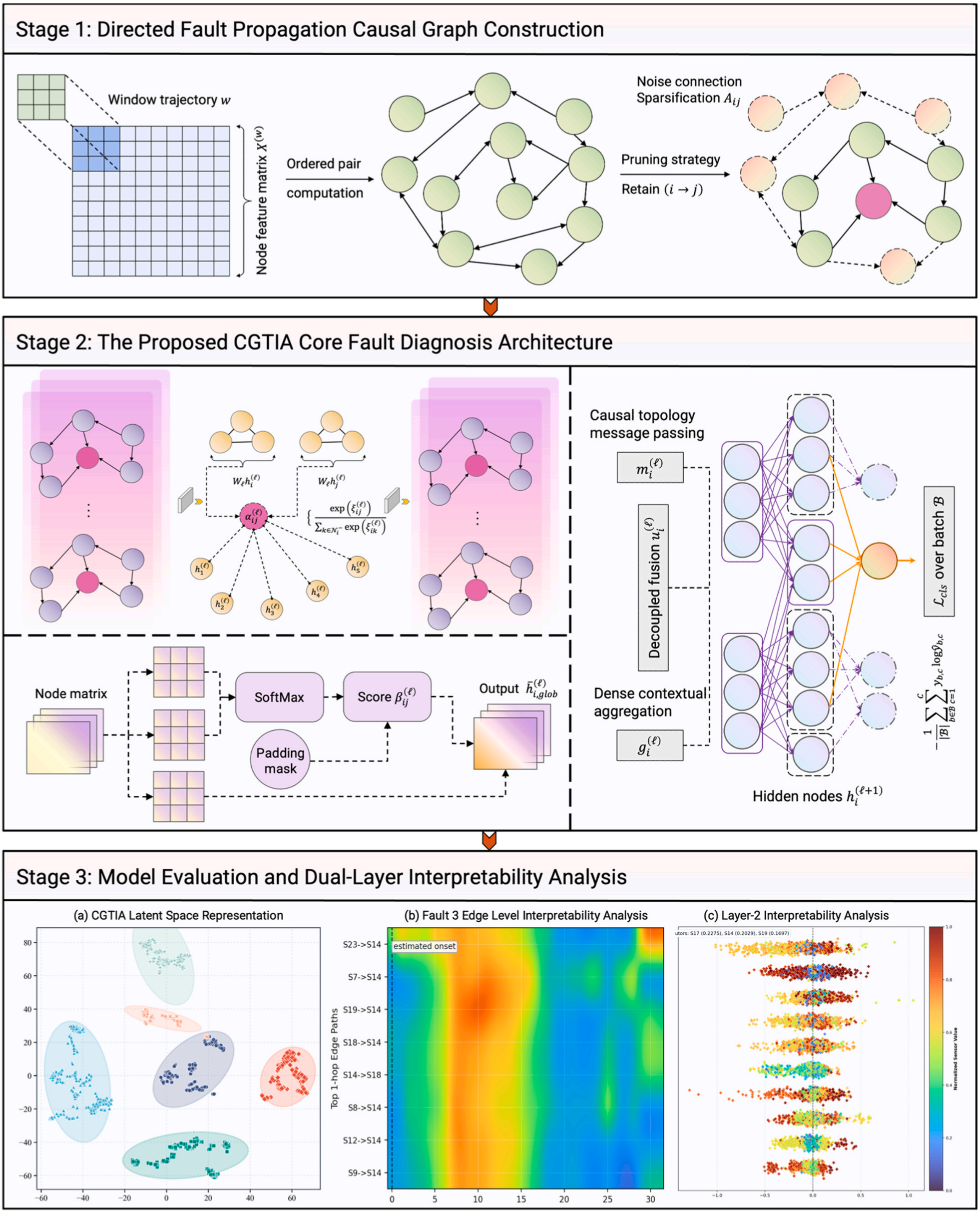


**Fig. 1.** The proposed end-to-end causal context graph transformer pipeline CGTIA for industrial process fault diagnosis.

$$x_j(t) = \sum_{\ell=1}^{L} a_\ell x_j(t-\ell) + \varepsilon_r(t) \quad (3)$$

The autoregressive memory using only sensor $j$ is presented in the null model Eq. (3), serving to define the baseline prediction error under the assumption that $i$ provides no additional directional information to $j$. As shown in Eq. (4), the lagged term from sensor $i$ extends the baseline model. The coefficient block models incremental cross-sensor effects across multiple delays, enabling the test to capture delayed propagation rather than merely synchronous coupling.

$$x_j(t) = \sum_{\ell=1}^{L} a_\ell x_j(t-\ell) + \sum_{\ell=1}^{L} b_\ell x_i(t-\ell) + \varepsilon_u(t) \quad (4)$$

The F-test statistic expressed in Eq. (5) quantifies whether the error reduction from Eq. (3) to Eq. (4) is substantial relative to model complexity, where $F_{i\to j}$ measures the significance of lagged sensor $i$ for predicting sensor $j$ at lag order $l$. A larger $F$ indicate stronger evidence that incorporating historical data from sensor $i$ yields non-trivial predictive gains for sensor $j$.

$$F_{i\to j} = \frac{(\mathrm{RSS}_r - \mathrm{RSS}_u)/L}{\mathrm{RSS}_u/(N-2L-1)} \quad (5)$$

The conventional significance decision can be written as:

$$\mathit{Retain}\ (i\to j) \Leftrightarrow p_{i\to j} < \alpha \quad (6)$$

Eq. (6) states the classical binary retention rule controlled by significance level $\alpha$, which determines whether the directed influence from sensor $i$ to sensor $j$ is retained. We adopt this rule as a statistical decision template, then move to a ranking-based construction to obtain a stable sparse graph under heterogeneous sensor dynamics.

During implementation, the causal strength derived from p-values is combined with node-wise pruning strategy to ensure robustness and controlled sparsity, as depicted in Fig. 2. Eq. (7) converts the multi-lag test results into a single continuous edge score. The minimum lag p-value emphasizes the strongest observed directional effects across delay periods, while the $-\log_{10}$ mapping converts small p-values into numerically separable positive strength values for ranking analysis.

$$S_{i\to j} = -\log_{10}\left(\min_{\ell=1,\dots,L} p_{i\to j}^{(\ell)}\right) \quad (7)$$

Node-wise sparsification is performed via Eq. (8), where each source node retains its top-k outbound causal links. In addition, a focused sensitivity analysis with respect to the top-k sparsification level was conducted to evaluate the robustness of the proposed framework, and the detailed results are presented in the Appendix. This avoids dense noise connections while preserving the directional influence strength of retained edges, yielding a graph complexity compatible with stable message passing.

$$A_{ij} = \begin{cases} S_{i\to j}, & j \in \mathrm{TopK}_i \\ 0, & \text{otherwise} \end{cases}, k = 6 \quad (8)$$

The above formula is equivalent to a node-adaptive threshold in Eq. (9), where $\tau_i$ denotes the adaptive threshold for source node $i$. Such thresholding is important since different sensors possess distinct causal radiation ranges, employing a single global threshold would bias the graph toward high-variance channels while suppressing still informative weak node connections.

$$j \in \mathrm{TopK}_i \Leftrightarrow S_{i\to j} \geq \tau_i \quad (9)$$

### 2.2. Graph transformer architecture for fault diagnosis

#### 2.2.1. Problem setup and representation

For each windowed sample $w$, the plant is represented as a directed sensor graph $G^{(w)} = (V, E, A)$, where $|V| = 24$ denotes the fixed set of sensing channels and $E$ is a directed edge set obtained from Granger screening. Each node corresponds to one sensor, and its input is not a static summary but a short trajectory within the window. To enable multi-branch reasoning on top of the trajectory input, a learnable projection is introduced to map the raw window vector into a latent channel space suitable for graph computation:

$$h_i^{(0)} = \phi_{\mathrm{in}}(x_i) = \mathrm{ELU}(W_{\mathrm{in}} x_i + b_{\mathrm{in}}) \quad (10)$$

First, it transforms heterogeneous sensor trajectories into a common representation space, enabling local and global branches to operate on embeddings with consistent scale and semantics. Second, it acts as a lightweight time compression layer, without requiring manually designed features (e.g., mean, variance, etc.), the projection learns which parts of the trajectory are informative for diagnosis.

#### 2.2.2. Local causal propagation branch

The local branch implements direction-sensitive message passing constrained by the directed Granger graph, as illustrated in Fig. 3. Our goal is to explicitly model how faults propagate through physically meaningful coupling paths, rather than allowing the architecture to discover adjacencies purely based on data correlations. Accordingly, the neighborhood update at layer $\ell$ is defined over the incoming neighbors of node $i$:

$$\tilde{h}_i^{(\ell)} = \mathscr{F}_\ell\left(h_i^{(\ell)}, \left\{h_j^{(\ell)}\right\}_{j\in\mathscr{N}_i^-}, \{e_{ij}\}_{j\in\mathscr{N}_i^-}\right) \quad (11)$$

Here $\mathscr{N}_i^-$ denotes in-neighbors of node $i$. A key architectural decision in the implementation is that the structural input for this branch contains only *edge_index*, explicit edge attributes $e_{ij}$ (e.g., Granger strength) are not consumed in forward computation. This is a deliberate design aimed at avoiding overparameterization of edge effects when causal evidence may be noisy or regime dependent. Instead, causal information enters

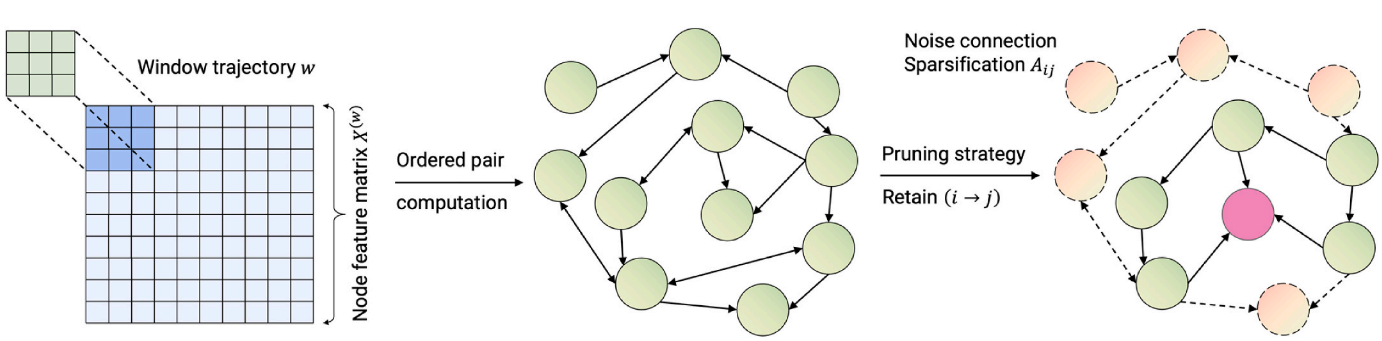


**Fig. 2.** Node sparsification of Granger causality for robust directed fault propagation graphs.

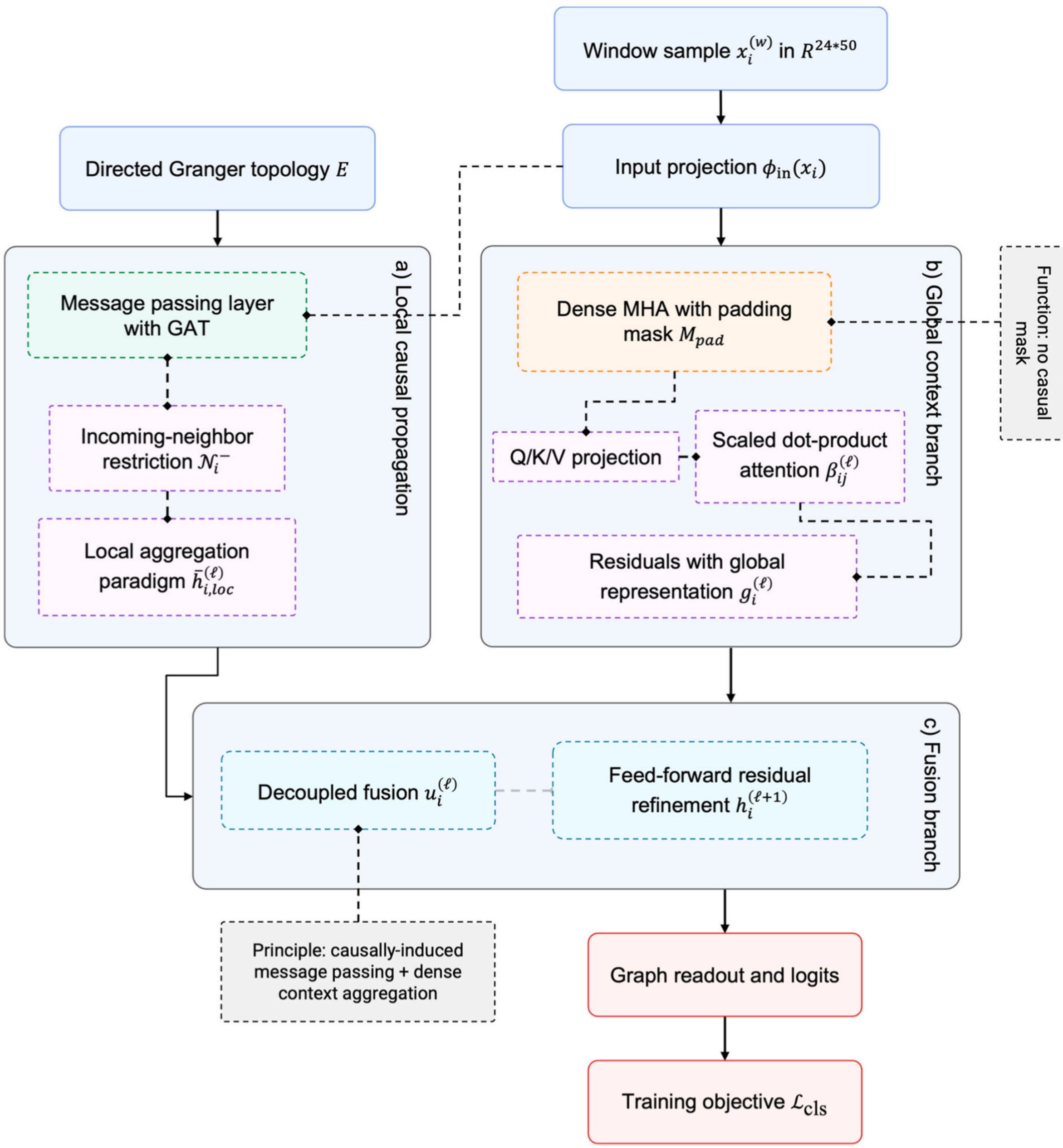


**Fig. 3.** The decoupled causal-guided graph architecture for multivariate process fault diagnosis.

$$\beta_{ij}^{(\ell)} = \text{softmax}_j\left(\frac{q_i^{(\ell)}\left(k_j^{(\ell)}\right)^\top}{\sqrt{d_k}} + M_{ij}\right) \tag{17}$$

primarily as a topological inductive bias that restricts where messages can flow, while allowing the model to learn how information aggregates along these paths.

To instantiate $\mathscr{F}_\ell$, we adopt GAT as the local operator (Brody et al., 2021). Attention logits and normalized coefficients are computed over $\mathscr{N}_i^-$:

$$\xi_{ij}^{(\ell)} = \text{LeakyReLU}\left(a_\ell^\top\left[W_\ell h_i^{(\ell)} \,\middle\|\, W_\ell h_j^{(\ell)}\right]\right), j \in \mathscr{N}_i^- \tag{12}$$

$$\alpha_{ij}^{(\ell)} = \frac{\exp\left(\xi_{ij}^{(\ell)}\right)}{\sum_{k\in\mathscr{N}_i^-}\exp\left(\xi_{ik}^{(\ell)}\right)} \tag{13}$$

$$\bar{h}_{i,loc}^{(\ell)} = \sum_{j\in\mathscr{N}_i^-}\alpha_{ij}^{(\ell)} W_\ell h_j^{(\ell)} \tag{14}$$

The above equations express a directed, localized aggregation process: node $i$ selectively integrates signals from its causal parents $j$, prioritizing upstream nodes whose embeddings align most closely with the current diagnostic context. It is important to emphasize that although GAT employs an attention mechanism, we treat it here as a learnable neighbor weighting mechanism rather than an interpretable signal (Rampášek et al., 2022).

Finally, the residual learning and normalization are applied to stabilize training and preserve information across layers:

$$m_i^{(\ell)} = \mathrm{Norm}_1\left(h_i^{(\ell)} + \mathrm{Drop}\left(\mathrm{LocalMPNN}\left(\overline{h}_{i,loc}^{(\ell)}\right)\right)\right) \tag{15}$$

### 2.2.3. Global context branch

Despite local propagation capturing directed interactions constrained by Granger topology, industrial processes also exhibit long-range and cross-unit phenomena. Such coupling often cannot be fully characterized by one hop causal chains within a short time window. To resolve this, we introduce a global context branch based on graph-wise dense self-attention (Vaswani et al., 2017). This branch is not intended to represent causal relationships but rather to provide a complementary mechanism for system-level context aggregation.

Given node embedding matrix $H^{(\ell)}$ at layer $\ell$, we compute query, key and value projections:

$$Q^{(\ell)} = H^{(\ell)}W_Q, K^{(\ell)} = H^{(\ell)}W_K, V^{(\ell)} = H^{(\ell)}W_V \tag{16}$$

Attention weights are then obtained by scaled dot-product attention. Here, $M_{ij}$ is a padding mask used only for batched execution (valid-node pairs use $M_{ij} = 0$). Critically, such design choice is central to the fault diagnosis viewpoint: the global branch is responsible for capturing distributed context — including shared operating conditions, global disturbances, or weak but consistent co-variations — that can support classification even when the local causal graph is sparse.

The global representation is as follows:

$$\overline{h}_{i,glob}^{(\ell)} = \sum_{j\in V}\beta_{ij}^{(\ell)}v_j^{(\ell)}, g_i^{(\ell)} = \mathrm{Norm}_2\left(h_i^{(\ell)} + \mathrm{Drop}\left(\overline{h}_{i,glob}^{(\ell)}\right)\right) \tag{18}$$

The residual form corresponds to the local branch but differs semantically, where $g_i^{(\ell)}$ is not causal propagation but context enriched, enabling node embeddings to reflect the broader system states.

### 2.2.4. Decoupled fusion mechanism

Local and global branches are fused additively, followed by a feed-forward residual refinement:

$$u_i^{(\ell)} = m_i^{(\ell)} + g_i^{(\ell)} \tag{19}$$

$$h_i^{(\ell+1)} = \mathrm{Norm}_3\left(u_i^{(\ell)} + \mathrm{MLP}\left(u_i^{(\ell)}\right)\right) \tag{20}$$

By fusing only after each branch has independently produced its own normalized output, we avoid conflating causal topology-induced message passing (Eq. (12)–(15)) with dense contextual aggregation (Eq. (16)–(18)). The separation is particularly crucial for fault diagnosis: local propagation aligns with the mechanistic intuition about fault spread, while global context captures nonlocal coupling and operating states.

After $L$ layers, graph-level embedding is computed by mean pooling:

$$h_G = \frac{1}{|V|}\sum_{i\in V}h_i^{(L)} \tag{21}$$

$$\widehat{y} = \mathrm{softmax}(W_c h_G + b_c) \tag{22}$$

Lastly, the supervised objective is cross-entropy over batch $\mathscr{B}$ shown in Eq. (23). In addition to promoting accurate fault diagnosis, the encoder learning process enables consistent representation across various operating modes and fault evolution stages.

$$\mathscr{L}_{cls} = -\frac{1}{|\mathscr{B}|}\sum_{b\in\mathscr{B}}\sum_{c=1}^{C}y_{b,c}\log\widehat{y}_{b,c} \tag{23}$$

## 2.3. Dual-layer interpretability module

The core layer-1 definition in this work is presented in Eq. (24). For each directed edge $(i,j)$, instead of reading the internal attention scores within graph transformer architecture, we compute endpoint sensitivities via gradient inputs and average both endpoints to obtain edge-level significance. This explicitly links edge importance to the output sensitivity of the predicted class $c^*$, while preserving the directed causal support $E$ from the Granger part.

$$e_{ij}^{(w)} = \frac{1}{2}\left(\frac{1}{T}\sum_{t=1}^{T}\left|\frac{\partial z_{c^*}^{(w)}}{\partial x_{i,t}^{(w)}}x_{i,t}^{(w)}\right| + \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\partial z_{c^*}^{(w)}}{\partial x_{j,t}^{(w)}}x_{j,t}^{(w)}\right|\right), (i,j)\in E \tag{24}$$

To achieve compact representation in layer-2, the node attribution magnitude is defined in Eq. (25). It compresses each sensor contribution within time window $w$ into a single non-negative value, enabling direct comparison across sensors and consistent propagation to both edge-level and global-level analysis. At the end, class-wise averaging and temporal binning on $e_{ij}^{(w)}$ produce layer-1 fault heatmaps that visually highlight sensor paths are emphasized during fault evolution.

$$a_i^{(w)} = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\partial z_{c^*}^{(w)}}{\partial x_{i,t}^{(w)}}x_{i,t}^{(w)}\right| \tag{25}$$

The attribution sign is retained in Eq. (26), therefore encoding directionality of influence: positive values indicate that sensor $i$ tends to increase the predicted logit for $c^*$, while negative values denote suppressive influence. This sign quantity complements Eq. (25), which captures only the magnitude of the effect.

$$s_i^{(w)} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\partial z_{c^*}^{(w)}}{\partial x_{i,t}^{(w)}}x_{i,t}^{(w)}\right) \tag{26}$$

Finally, global sensor importance is shown as follows:

$$I_i = \frac{1}{|\mathscr{W}|}\sum_{w\in\mathscr{W}}a_i^{(w)} \tag{27}$$

Here, sensor importance at the dataset level is defined by averaging Eq. (25) over the evaluation window $\mathscr{W}$. The aggregation method stabilizes fluctuations at the sample level and generates a global ranking suitable for Top $N$ reporting. In our framework, this ranking is interpreted as a process-level indicator reflecting which sensors consistently support fault identification across different operating conditions.

# 3. Results and discussion

## 3.1. Cranfield multiphase flow benchmark

The experimental data used in this study are acquired from the multiphase flow facility at Cranfield University (Ruiz-Cárcel et al., 2015), a pilot-scale test rig established under industrial operating conditions to support research on separation processes and operational monitoring. The apparatus manages controlled mixtures of air and water through a pressurized pipeline-riser-separator network, simulating fundamental dynamics encountered in upstream oil and gas production systems (Fan et al., 2023; Sun et al., 2022). As shown in Fig. 4, the facility comprises four subsystems: a fluid supply section (including air compressors, water pump and associated control valves), a 4-inch pipeline extending approximately 55 m to a 10.5 m catenary riser, a two-phase top separator at the riser outlet, and an 11 m$^3$ three-phase separator at ground level. A parallel 2-inch pipeline and 4-inch bypass pipeline are available but remain isolated during normal operation. Compared to the conventional simulated diagnostic benchmark TEP, Cranfield system uniquely incorporates real process noise, nonlinear multiphase flow dynamics and coupling between fault effects and time-varying conditions. Throughout both normal and fault modes, flow setpoints are purposefully adjusted to challenge fault diagnosis algorithms in distinguishing fault-induced deviations from routine operational transients.

The entire plant is managed via the DeltaV supervisory control network, capturing all process variables at a unified sampling rate. As shown in Table 1, six fault features are introduced in the facility, each

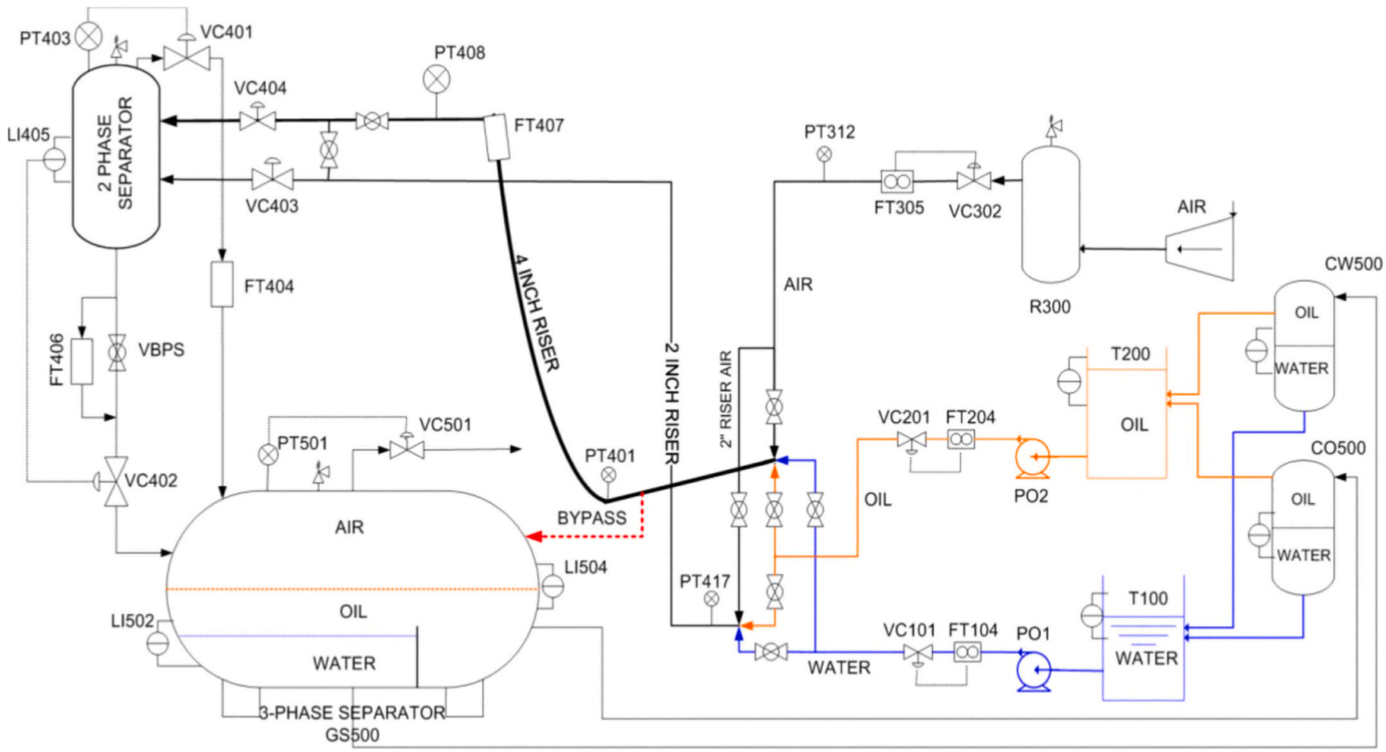


**Fig. 4.** The schematic diagram of the Cranfield multiphase flow layout.

**Table 1**
An overview of fault scenarios in the Cranfield multiphase flow benchmark.

| No. | Description | Type | Mechanism |
|---|---|---|---|
| 1 | Air line blockage | Incipient | Valve closure |
| 2 | Water line blockage | Incipient | Valve closure |
| 3 | Top separator inlet blockage | Incipient | VC404 closure |
| 4 | Open direct bypass | Incipient | Bypass valve |
| 5 | Slugging conditions | Intermittent | Low flow rates |
| 6 | 2-inch line pressurization | Abrupt | Bridge valve |

intended to simulate real-world situations in multiphase flow systems. Faults 1 through 4 are incipient in nature, introducing gradually developing blockages or bypass leaks through incremental valve manipulation to observe detection sensitivity as fault severity increases. Fault 5 involves intermittent slugging triggered by liquid accumulation and blowing through within the riser caused by reduced flow velocities. Fault 6 represents an abrupt operational anomaly caused by mistakenly pressurizing the normally isolated 2-inch pipeline, and additional sensors (PT417, S24) are specifically installed to monitor this fault. Furthermore, a total of 24 process observations is observed, as outlined in Table 2. It is noteworthy that multiple variables share the same physical instrumentation: the flowmeter FT407 at the top of the riser provides S10 (flow rate), S13 (density) and S16 (temperature); FT406 at the separator outlet provides S12, S14 and S17, while the feedwater flowmeter FT104 provides S9, S15 and S18. This co-location creates inherent correlations, demanding that proposed methods must be capable of distinguishing them from fault-induced dependencies.

### 3.2. Model training and hyperparameter configuration

The CGTIA training pipeline couples a fixed directed Granger prior with the graph transformer classifier optimized on windowed sensor graphs. The graph is precomputed offline using pairwise causality scoring with maximum lag $L = 5$ and top 6 outgoing neighbors per sensor, producing a sparse directed topology with 24 nodes. Feature samples are generated at a 1 Hz sampling rate with 50s time windows and 1s strides. Global normalization is applied during preprocessing as shown in Eq. (28), where $x_{i,t}$ denotes the original reading of sensor $i$ at time $t$. The special case $d_i = 1$ avoids division by zero when the sensor is constant; in this instance, the sensor uses $\widehat{x}_{i,t} = 0$ at all positions. The dataset comprises 92,149 time windows, stratified into training, validation and test at a 70:15:15 ratio, resulting in 64,504, 13,822 and 13,823 sample sets respectively. The training configuration and hyperparameters are optimized by Adam under multiclass cross-entropy loss. In short, this training section therefore focuses on data preparation, hyperparameters and convergence behavior for the architecture defined in Section 2.2 (Fig. 3).

$$\widehat{x}_{i,t} = \frac{x_{i,t} - m_i}{d_i}, d_i = \begin{cases} M_i - m_i, \text{if } M_i \neq m_i \\ 1, \text{if } M_i = m_i \end{cases} \tag{28}$$

The loss trajectory for proposed architecture is shown in Fig. 5. During the initial stage (epochs 1-3), both training and validation losses exhibit steep declines, signifying efficient gradient-driven adaptation guided by causal structural priors. In the intermediate period (epochs 4-8), the generalization gap within the shaded region between curves remains consistently narrow and controlled, and validation accuracy peaks at 0.9933 b y epoch 8. Throughout the late stage, validation loss fluctuates only slightly after reaching its minimum of 0.0195 in the 9th cycle, while training loss continues to decrease marginally as the cosine annealing learning rate approached zero. The final test evaluation yields a loss of 0.0207 and an accuracy of 0.9925, both highly aligned with the optimal validation results. In short, the convergence behavior highlights two critical aspects: First, the sustained narrow generalization gap suggests that our constructed topology, functioning as an effective structural regularizer, constraints the CGTIA hypothesis space to causally plausible sensor interactions. Furthermore, the high consistency between validation and test metrics suggests the proposed method can stabilize at high performance within the short epoch schedule without requiring extended training time.

**Table 2**
All process observations and subsystem types in the multiphase flow benchmark.

| Sensor_ID | Location | Measured variable | Subsystem | Type |
|---|---|---|---|---|
| S1 | PT312 | Air delivery pressure | Air supply | Pressure |
| S2 | PT401 | Pressure at riser bottom | Riser | Pressure |
| S3 | PT408 | Pressure at riser top | Riser | Pressure |
| S4 | PT403 | Pressure in top separator | Top separator | Pressure |
| S5 | PT501 | Pressure in three-phase separator | Three-phase separator | Pressure |
| S6 | PT408 | Differential pressure (riser) | Riser | Derived |
| S7 | PT403 | Differential pressure over VC404 | Top separator inlet | Derived |
| S8 | FT305 | Air flow rate (input) | Air supply | Flow rate |
| S9 | FT104 | Water flow rate (input) | Water supply | Flow rate |
| S10 | FT407 | Flow rate at riser top | Riser | Flow rate |
| S11 | LI405 | Liquid level in top separator | Top separator | Level |
| S12 | FT406 | Flow rate at top separator outlet | Top separator | Flow rate |
| S13 | FT407 | Fluid density at riser top | Riser | Density |
| S14 | FT406 | Fluid density at top separator outlet | Top separator | Density |
| S15 | FT104 | Water input density | Water supply | Density |
| S16 | FT407 | Temperature at riser top | Riser | Temperature |
| S17 | FT406 | Temperature at top separator outlet | Top separator | Temperature |
| S18 | FT104 | Temperature of water input | Water supply | Temperature |
| S19 | LI504 | Gas-liquid level in three-phase separator | Three-phase separator | Level |
| S20 | VC501 | Position of valve VC501 | Three-phase separator | Valve |
| S21 | VC302 | Position of valve VC302 (air) | Air supply | Valve |
| S22 | VC101 | Position of valve VC101 (water) | Water supply | Valve |
| S23 | PO1 | Water pump current | Water supply | Electrical |
| S24 | PT417 | Pressure in 2-inch line (mixture zone) | 2-inch flow line | Pressure |

### 3.3. Fault diagnosis performance and comparison

#### 3.3.1. Diagnostic confusion matrix

Fig. 6 displays the confusion matrix of proposed CGTIA on Cranfield test set, presenting a complete picture of the model behavior characteristics across various categories. The model achieves an overall test accuracy of 99.25% on 13,823 test samples, correctly classifying 13,719 instances. More importantly, category-specific breakdown reveals this performance is not concentrated in a few dominant types. Fault 3 (top separator inlet blockage, 3897/3897) and fault 6 (2-inch pipeline pressurization, 1129/1129) are classified without a single error. Fault 2 (water line blockage) and fault 4 (open direct bypass) demonstrated virtually perfect recognition, with only 2 and 4 falsely detected samples respectively. Fault 5 (slugging conditions) exhibited 10 misclassifications out of 1958 test samples, corresponding to a category recall of 99.49%.

The only significant deviation from near-perfect performance occurs for fault 1 (air line blockage), where 88 out of 2168 samples are detected as fault 6, reducing recall for this category to 95.94%. These confused samples accounted for 84.6% of all 104 errors in the test set, becoming the decisive indicator of the CGTIA erroneous features. The confusion exhibited strict directionality: all 88 errors flowed from F1 to F6, with zero instances of reverse diagnosis. Such asymmetry holds critical implications, demonstrating that the model maintains robust and unambiguous internal representations for line pressurization. Conversely, the feature space for F1 contains a boundary region where features may overlap with those of F6 under specific operating conditions. The remaining 16 incorrect predictions fall into three minor patterns: 7 samples of F5 are misclassified as F2 (slugging confused with water blockage), 3 samples of F5 are wrongly identified as F4 (slugging confused with bypass), and 4 samples of F4 are scattered across faults 1, 2 and 3.

From a process safety perspective, these results indicate that the proposed CGTIA framework can serve as a highly reliable diagnostic support method under realistic disturbances, but they also highlight that even rare misclassifications deserve explicit consideration in deployment. In particular, the observed asymmetric confusion from F1 to F6 suggests that, under certain operating conditions, air-line blockage may

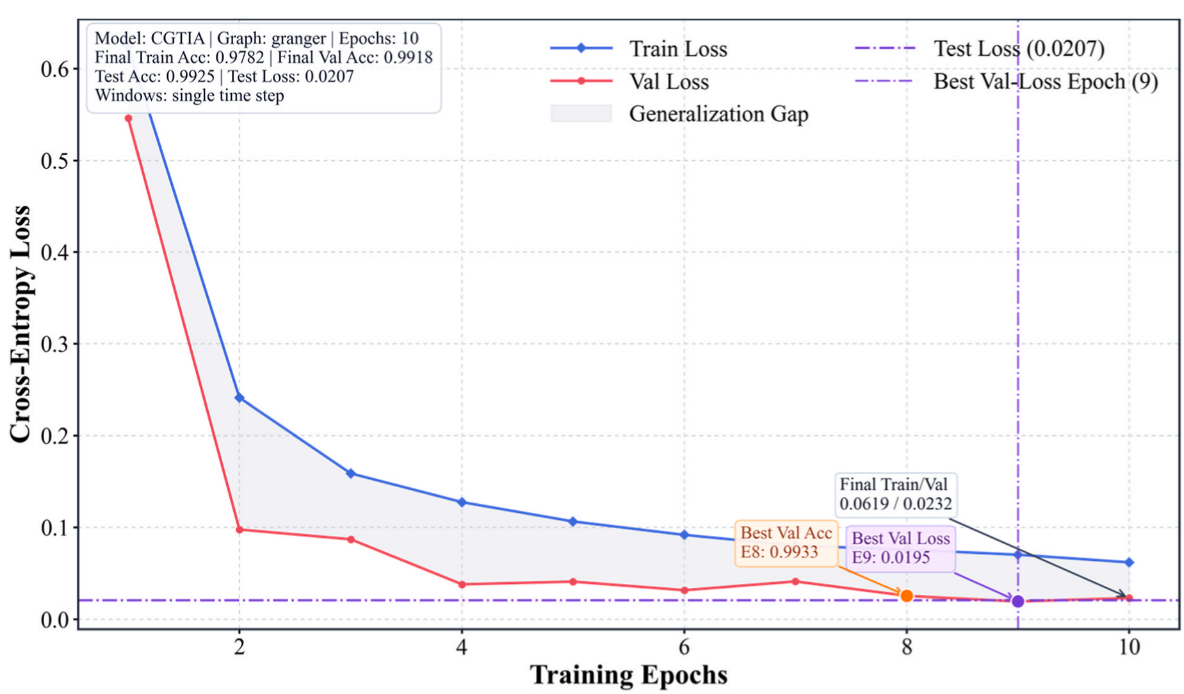


**Fig. 5.** Training and validation loss curves for the proposed CGTIA framework over 10 epochs.

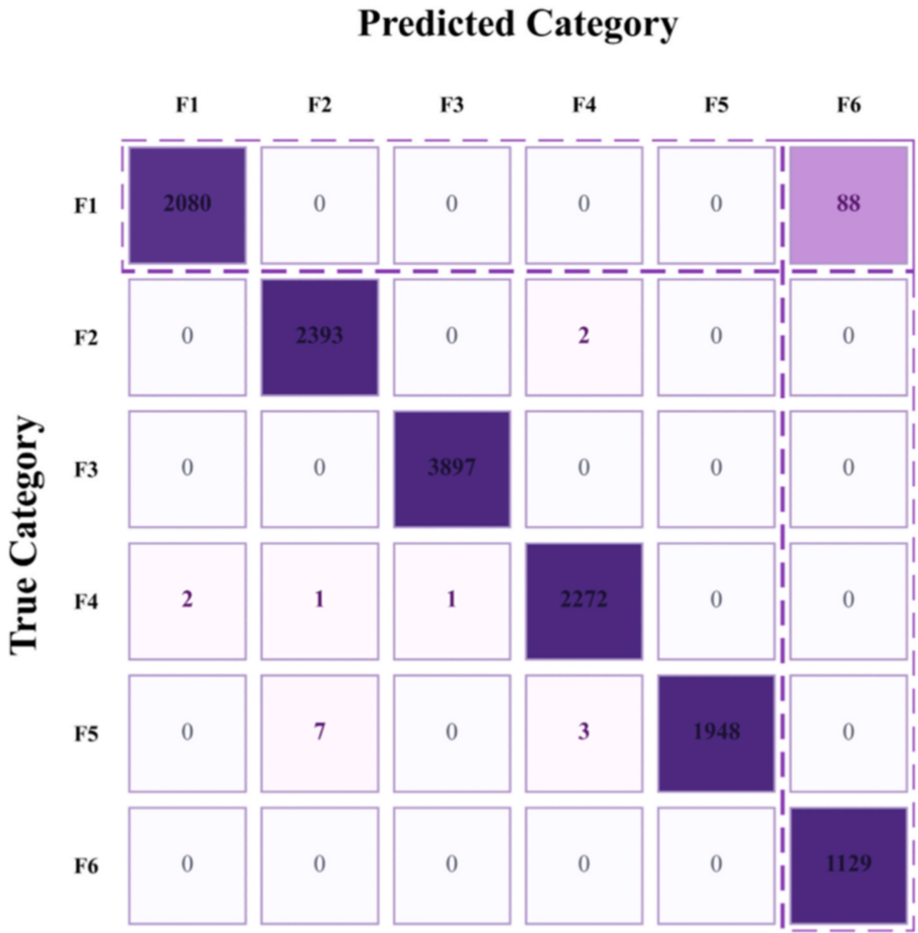


**Fig. 6.** The confusion matrix of proposed CGTIA on Cranfield benchmark (off-diagonal entries represent misclassification).

exhibit features partially resembling line pressurization. In a real plant, such confusion could influence alarm prioritization, operator interpretation of abnormal conditions, or the selection of follow-up inspection actions. Therefore, although the overall error rate remains very low, model predictions should be integrated with existing alarm logic, process knowledge and operator verification procedures rather than being treated as fully autonomous safety decisions.

### 3.3.2. *Latent representation performance*

To understand how the CGTIA architecture transforms raw inputs into discriminative representations, Fig. 7 compares t-Distributed Stochastic Neighbor Embedding (t-SNE) projections of the learned latent space and the original windowed feature space, both colored by fault category. In the raw feature space, the six fault classes display substantial overlap. F1 and F2 are intermingled in the upper extended region of the projection, while F4 and F5 are distributed along an intersecting diagonal band. Samples of F6 are scattered across multiple

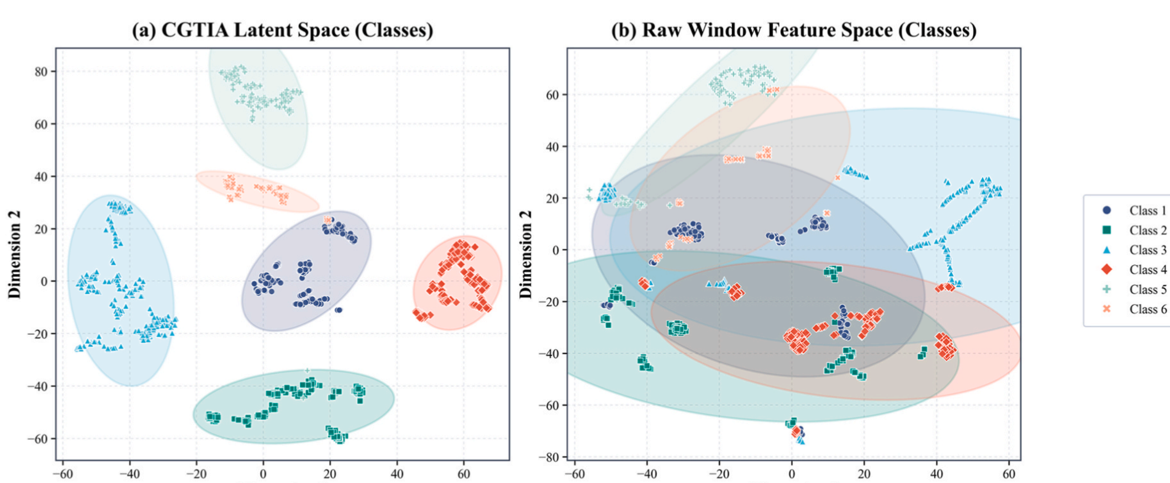


**Fig. 7.** T-SNE visualization of the proposed method: (a) the latent representations learned by CGTIA and (b) the raw windowed feature space.

disjoint clusters, failing to form coherent groupings. The overlapping distribution is expected, as raw sensor readings at any given moment are dominated by current flow setpoints rather than fault characteristics. The core diagnostic challenge is precisely to extract fault-specific components from such operation-dominated feature spaces.

In the CGTIA latent space, the separation between classes is markedly improved. Each fault type forms compact, well-isolated clusters with clearly discernible boundaries. Fault 3 with green triangle and fault 6 with orange cross constitute the cohesive clusters, consistent with the perfect recall in the test set. Fault 5 is distinctly separated into an independent region, reflecting the model's ability to represent the distinctive oscillatory dynamics of slugging as a coherent latent structure. Crucially, fault 1 and fault 6 partially overlapping in the raw space evolve into distinct clusters separated by a narrow gap in the latent space. The existence of such overlap directly corroborates the previous confusion matrix finding, where 88 misclassified F1 samples correspond to boundary instances in the transition zone between the two clusters.

### 3.3.3. *Comparative model evaluation*

After analyzing the diagnostic features of CGTIA, we evaluate its performance against five baseline architectures. Fig. 8 summarizes various statistical metrics in a radar chart and shows that proposed approach achieves the most balanced performance across criteria, yielding the highest composite score of 0.9911, ahead of ChebNet (J. Liu et al., 2025), MixHop (Zhuang et al., 2024), GraphSAGE (Seon et al., 2023), GCN (Yang et al., 2022), and GAT (Wu and Zhao, 2023). One instructive contrast here lies between proposed method and GAT, where they differ by >26% in composite scores. The distinction is that the standard GAT ignores temporal priority and causal direction when distributing attention, whereas CGTIA constraints local message passing branches to follow directed causality edges. Three intermediate baselines obtain composite scores ranging from 0.89 to 0.95, indicating that even without explicit causal constraints, graph structures provide meaningful inductive biases for fault diagnosis. However, despite employing advanced neighborhood sampling strategy, GraphSAGE exhibits lower recall than CGTIA, suggesting that random subgraph sampling cannot capture directed causal relationships as effectively as the complete edge set based on Granger causality.

The left panel of Fig. 9 refines the comparison to the per-fault level. CGTIA achieves a recall ranging from 95.94% (F1) to 100% (F3 and F6) for all faults, consistently outperforming all baseline models. In F5 scenario, GraphSAGE reaches only approximately 78% recall, GCN drops to about 58%, while CGTIA still attains 99.49%. The improvement stems from our constructed causal graph encoding temporal sequences of pressure buildup, slug formation and downstream level oscillation. Such dynamic characteristics remain uncaptured by non-causal architectures like GCN and standard GAT. For water line blockage, the recall of GCN drops to approximately 66% while CGTIA maintains 99.92%, confirming the distinct advantage of proposed method when faults produce subtle and temporally extended characteristics.

The right side further illustrates the trend of all models ranked by comprehensive performance through macro recall and macro error rate. The macro recall curve rises steeply from GAT (67.48%) through GCN (69.55%) to GraphSAGE (87.27%), then flattens near MixHop (93.30%) and ChebNet (93.70%) before peaking at 99.20% for CGTIA. The macro error rate correspondingly decreases from 32.52% of GAT to 0.80% of CGTIA. The transition from the steep improvement region (GAT to GraphSAGE) to the plateau (MixHop to ChebNet) suggests that as models fully learn the graph topology, more complex graph processing yields diminishing returns. Notably, the CGTIA breaks through the plateau by expressly incorporating causal directionality, reducing macro error rate by an order of magnitude lower than the plateau baselines of approximately 6%.

Lastly, Table 3 provides supplementary diagnostic insights via additional statistical metrics including precision and F1-score. The CGTIA achieves classification precision exceeding 99.7% across all six

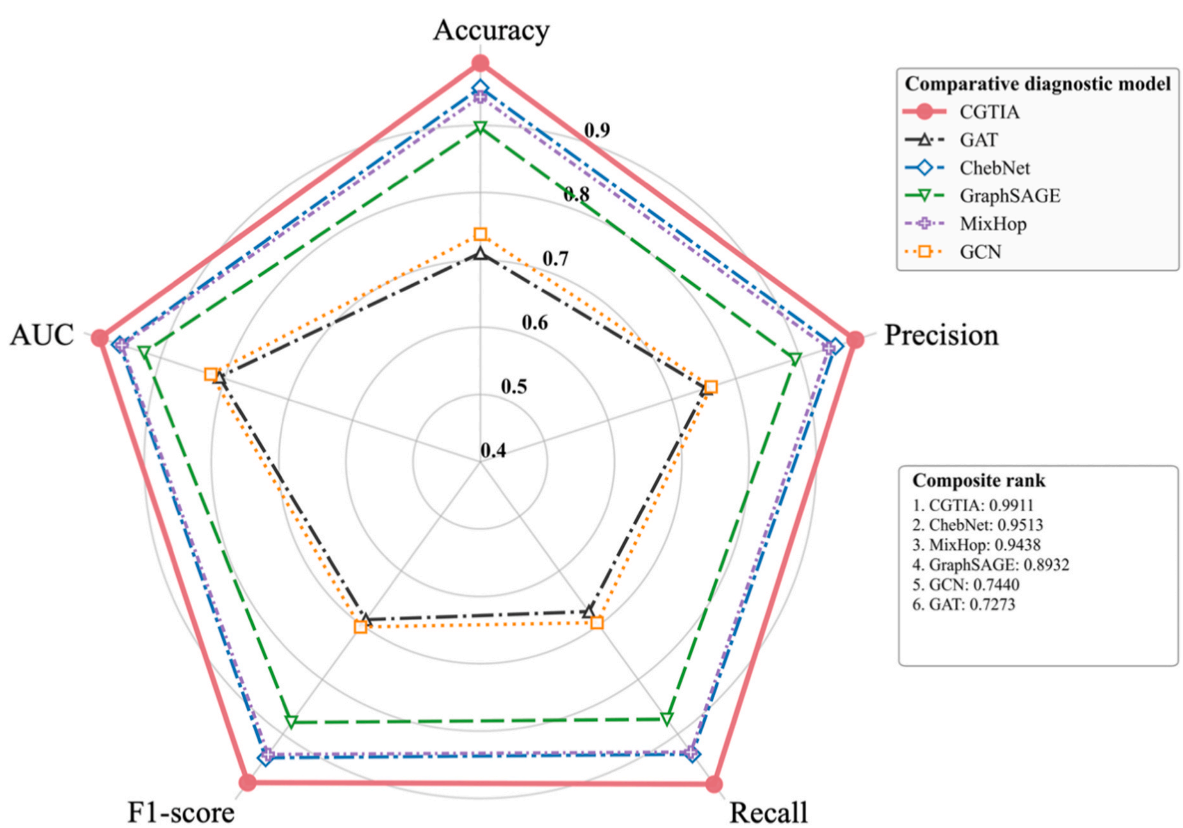


**Fig. 8.** The radar chart comparing CGTIA with five leading baseline models across various statistical metrics (accuracy, precision, recall, F1-score, AUC) on Cranfield benchmark.

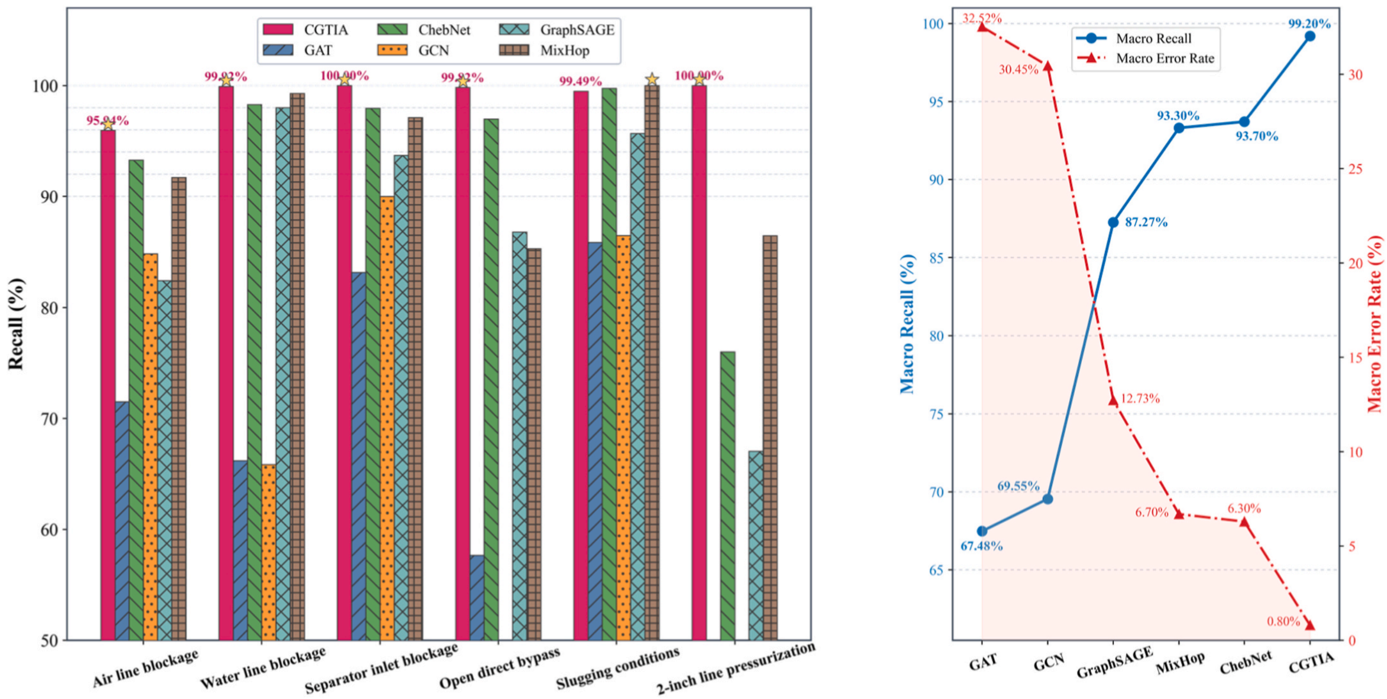


**Fig. 9.** Left: Per-fault recall comparison with CGTIA values annotated above each bar. Right: Trend plot of macro recall and error rate trend sorted by overall performance.

**Table 3**
Precision (Pr, %) and F1-score (Score, %) for CGTIA versus baseline models.

| Model | Fault 1 | | Fault 2 | | Fault 3 | | Fault 4 | | Fault 5 | | Fault 6 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pr | Score | Pr | Score | Pr | Score | Pr | Score | Pr | Score | Pr | Score |
| **CGTIA** | 99.90 | 97.88 | 99.67 | 99.79 | 99.97 | 99.99 | 99.78 | 99.80 | 100.0 | 99.74 | 92.77 | 96.25 |
| GAT | 53.93 | 61.48 | 49.75 | 56.80 | 85.88 | 84.50 | 81.19 | 67.42 | 89.08 | 87.44 | 92.09 | 56.31 |
| ChebNet | 88.65 | 90.90 | 95.61 | 96.93 | 99.43 | 98.68 | 92.73 | 94.80 | 97.75 | 98.74 | 99.42 | 86.14 |
| GCN | 60.43 | 70.58 | 68.51 | 67.15 | 93.47 | 91.70 | 50.53 | 50.24 | 83.73 | 85.08 | 100.0 | 57.36 |
| GraphSAGE | 86.58 | 84.45 | 81.61 | 89.05 | 97.78 | 95.69 | 84.91 | 85.83 | 94.26 | 94.96 | 90.55 | 77.05 |
| MixHop | 89.87 | 90.78 | 86.28 | 92.33 | 99.42 | 98.25 | 91.90 | 88.47 | 99.85 | 99.92 | 100.0 | 92.73 |

faults, with F1-scores ranging from 96.25% to 99.99%. The ChebNet model, ranked second overall, reveals its primary limitation on F6 with F1 score of 86.14%. The decline in precision is attributed to the same confusion between F1 and F6 previously noted, confirming this pattern is a dataset-level difficulty rooted in the physical similarity of two fault features while not a deficiency unique to any single model. MixHop delivers competitive F1-scores (exceeding 92%) on faults 3, 5 and 6, but underperforms on F2 with 92.33% and F4 with 91.90%, showing limitations in capturing bypass-related flow diversion dynamics. GAT and GCN exhibit the largest per-fault variations, with scores dropping to the 50-61% range on the weakest classes. This inconsistency renders them unsuitable for safety-critical industrial deployments, where diagnosis framework must maintain reliable performance against all fault types as a prerequisite.

### 3.4. Edge and node level interpretability analysis

In this section, we interpret CGTIA from both edge and node perspectives, linking fault-wise propagation paths to global sensor importance. Consider F6 as an example, which is triggered by opening the bridging valve between the 4-inch and 2-inch flow lines at the top of the connecting riser. The operation pressurizes the 2-inch pipeline, which is normally isolated. According to the benchmark reference, this fault is intended to simulate an abnormal system operating condition and should, in theory, not significantly affect the flow conditions in the main 4-inch pipeline. Therefore, diagnostic signals for F6 are largely confined to a single additional measurement: PT417 (S24), the 2-inch pipeline pressure value, which was only integrated into the analysis during this specific failure mode. With only 104 windows in the interpretability subset (the smallest among the six fault categories), F6 presents the most challenging scenario in terms of both classification and interpretability.

Despite these challenges, the edge-level heatmap in Fig. 10 demonstrates a highly consistent pattern for F6. Here, S24 (PT417) is properly identified as the primary causal driver, appearing in four of the top ranked edges: S24→S14, S24→S15, S24→S20 and S24→S22. The pattern indicates that CGTIA framework can trace secondary impacts from the physical fault source — the pressurized 2-inch pipe — to downstream effects on density, temperature and valve control subsystems. The S24→S22 (PT417→VC101, water valve position) edge is particularly noteworthy, showing the module captured the subtle coupling where pressurized 2-inch piping disrupts system hydraulic equilibrium, potentially triggering compensatory adjustments in the water control valve. In fact, the proposed CGTIA can correctly determine S24 as the crucial upstream variable using limited samples, demonstrating its ability to learn sufficiently effective structural prior information to guide attention even under limited data conditions. Furthermore, it is worth noting that all dominant edge paths and their physical interpretations are summarized in Table 4.

As a complement to the edge score analysis, Fig. 11 presents node-level interpretability results, consisting of global sensor importance

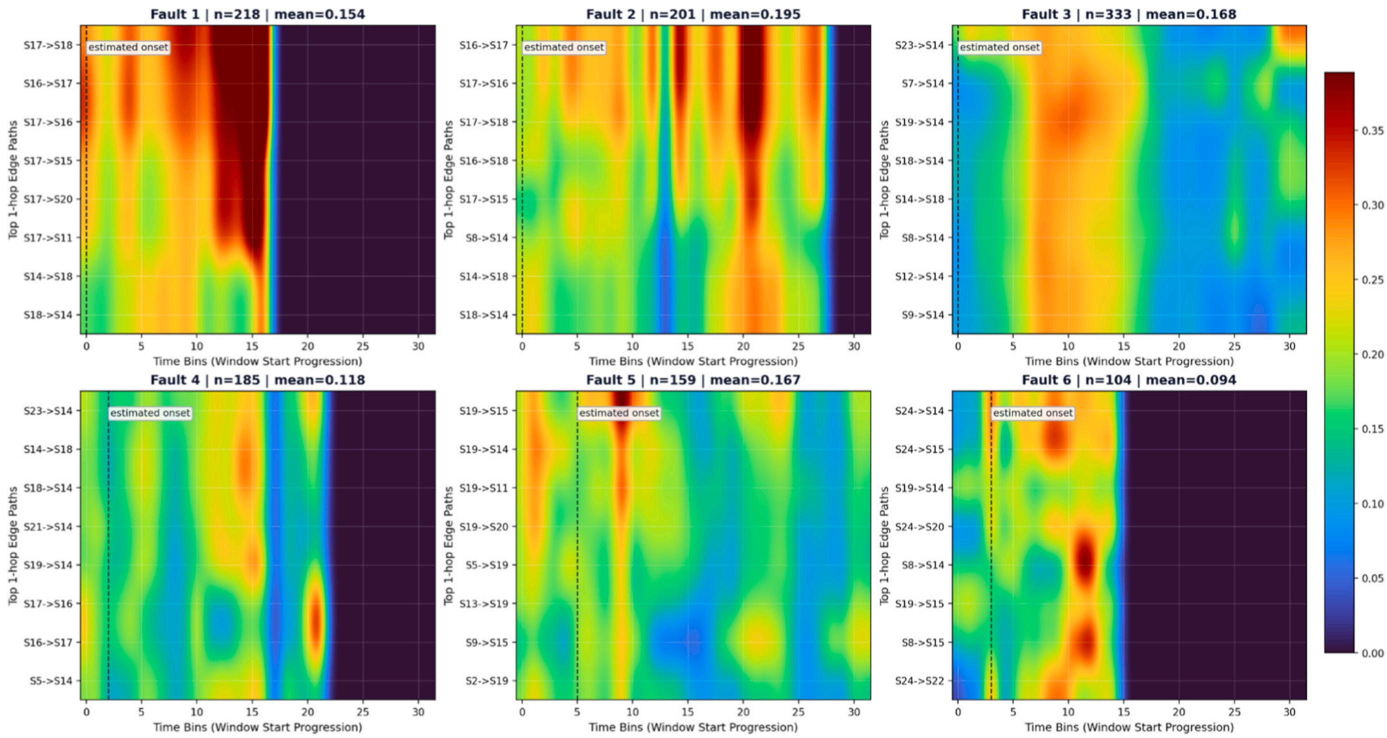


**Fig. 10.** Fault-wise edge score heatmaps on the top interpretability layer over Granger causality graph.

**Table 4**
A summary of dominant edge paths and physical interpretations.

| Fault_ID | Mean | Domain edges | Physical interpretation |
|---|---|---|---|
| F1 | 0.154 | S17→S18, S16→S17 | Air blockage alters the gas-liquid ratio, propagating through the temperature pathways at the riser and separator |
| F2 | 0.195 | S16→S17, S17→S18, S8→S14 | Water blockage reduces liquid flow, causing density changes at separator output (S14) with upstream air flow (S8) compensation |
| F3 | 0.168 | S23→S14, S7→S14, S19→S14 | VC404 blockage directly captured via differential pressure (S7); pump current (S23) and separator level (S19) converge on density hub (S14) |
| F4 | 0.118 | S23→S14, S14→S18, S21→S14 | Bypass diverts flow from riser; air valve position (S21) and pump current (S23) reflect compensatory control response |
| F5 | 0.167 | S19→S15, S19→S14, S19→S11 | Three-phase separator level (S19) acts as causal hub, receiving propagated pressure and density oscillations from slugging in the riser |
| F6 | 0.094 | S24→S14, S24→S15, S24→S20 | PT417 (S24) properly identified as fault source; pressurization propagates to density and valve position sensors |

rankings (left plot) and corresponding attribution distributions (right plot). The global importance ranking shows S17 (FT406, top separator outlet temperature) leading with a score of 0.2275, followed by S14 (FT406, top separator outlet density) at 0.2029, and S19 (LI504, three-phase separator liquid level) at 0.1697. The importance hierarchy displays a steep decay, with a gap of 0.129 between the top-ranked sensor and tenth-ranked S21 (VC302, air valve position, 0.0981). This indicates CGTIA can make sharp distinctions with the highest diagnostic value instead of allocating attention uniformly. Physically, the FT406, located at the bottom outlet of the top separator, jointly measures temperature (S17) and density (S14). The measurements effectively incorporate the conditions within the top separator, situated at the junction of the riser, bypass pipe and separator cascade. Consequently, the separator outlet naturally converges multiple fault characteristics, confirming why S17 and S14 rank as the top two globally in importance.

The attribution distribution map illustrates the sensor-level structure underlying these global scores. For S17, individual attribution values span a wide range from approximately 1.0 to +0.5, displaying a distinct color gradient. High normalized sensor values predominantly correspond to positive attributes, while low sensor values align with negative or near-zero attributions. The monotonic relationship between sensor values and attribution direction indicates that the model has established a physically coherent mapping relationship. For instance, elevated separator output temperatures signify process anomalies, aligning with the thermodynamic behavior observed during multiphase flow disruption. Also, S14 (density) exhibits a more concentrated cluster centered slightly below zero, with significant positive attribution outliers appearing in the high-value region. This pattern suggests S14 contributes most decisively under specific fault conditions generating abnormal density readings, rather than contributing equally across all samples.

Collectively, the edge-level analysis reveals which sensor pairs the model focuses on for each category, and node-level exploration offers a consolidated view across faults, highlighting the individual sensors that contribute most to the overall diagnostic capability. The consistency between the dominant paths in Table 4 and the global sensor rankings in Fig. 11 supports that model decisions are grounded in physically plausible process couplings rather than incidental co-variations. From a practical reliability viewpoint, these interpretability outputs may also support reliability-oriented actions beyond fault recognition itself. For instance, dominant edge paths can provide useful cues for fault isolation confidence and inspection prioritization, while globally important sensors may help guide targeted monitoring or maintenance planning toward the most diagnostically informative subsystems. In this way, CGTIA may serve not only as a fault diagnosis model, but also as a structured support tool for reliable decision-making in industrial process systems.

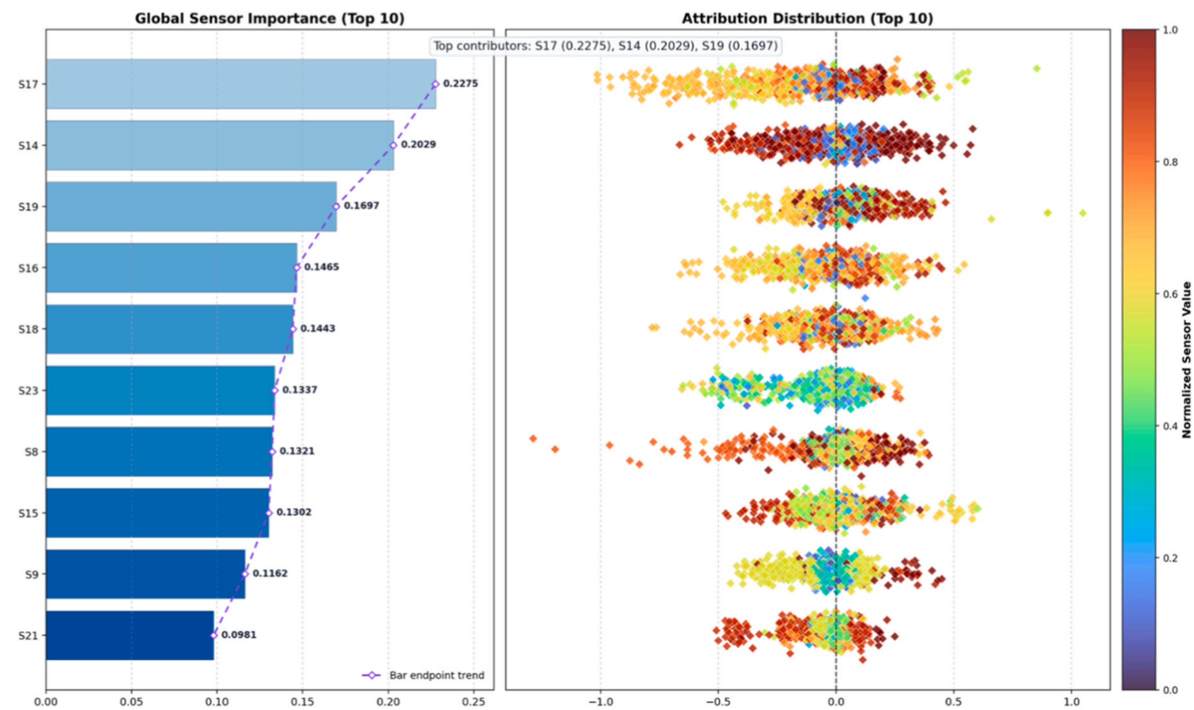


**Fig. 11.** Node-level interpretability analysis of proposed CGTIA: global sensor importance ranking (left) and sample attribution distribution (right).

## 4. Conclusions

This paper presents CGTIA, a causal-guided graph learning framework designed to improve both the reliability and interpretability of industrial fault diagnosis. The method constructs the window-level directed graph using lag-aware Granger evidence and enforces node-wise sparsification to obtain a robust, controllable propagation topology. On top of this prior, the decoupled architecture separates local causal propagation from global contextual aggregation, thereby reducing entanglement between directed propagation structure and dense covariance modeling. Beyond conventional simulation-centered validation, we evaluated the approach on a real multiphase flow facility dataset with pronounced class imbalance, demonstrating strong diagnostic capability under realistic fault rarity. In terms of Recall, the proposed method achieves the superior results (ranging from 95.94% to 100% for all faults) when compared against five baseline models. The interpretability study further yields mechanistic observations: for instance, F6 exhibits weak and localized signatures yet the model consistently traces dominant influence around PT417, while the layer-2 analysis highlights separator outlet sensing as globally informative across fault types. Overall, these results indicate that CGTIA can achieve high performance while producing explanations that align with physically plausible couplings in the process. At the same time, the present study should be interpreted within a defined scope. The directed prior is constructed from lag-aware Granger evidence and fixed during model training, which means that causal structure uncertainty and time-varying topology are not yet explicitly modeled. In addition, the current validation is performed on a representative real multiphase flow facility, and generalization across different industrial settings remains an important topic for following study.

In future work, several promising directions merit further exploration, particularly in response to the present framework's current limitations. First, we will investigate uncertainty-aware, adaptive causal graph construction to improve robustness when causal evidence is weak or contaminated by measurement noise. Second, we will integrate large language models (LLMs) to translate CGTIA's edge and node-level attributions into operator-oriented explanations, enabling interactive diagnostic narratives that remain faithful to the model directed propagation logic. To mitigate hallucinated content, we will incorporate retrieval grounding from historical logs, alarms and incident reports, so that generated explanations are supported by traceable evidence and consistent with the inferred propagation paths. Finally, we will develop human-in-the-loop decision support that proposes and iteratively refines verification actions (e.g., targeted sensor checks and subsystem inspections) while remaining constrained by the causal topology and CGTIA local to global decomposition, thereby improving practicality and trustworthiness for deployment in real industrial process monitoring.

## CRediT authorship contribution statement

**Shuaiyu Zhao:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoyu Yang:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Zihao Wang:** Writing – review & editing, Validation, Methodology, Investigation. **Faisal Khan:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation. **Qingsheng Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Given his role as Editor, Dr. Qingsheng Wang had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

org/10.1016/j.jlp.2026.106022.

## Data availability

Data will be made available on request.

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